

BHUJ, India

WMO: 426340

Lat: 23.288N

Lon: 69.670E

Elev: 82

StdP: 100.35

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.3	(c) 11.2	(d) -5.1	(e) 2.5	(f) 20.9	(g) -2.3	(h) 3.1	(i) 21.3	(j) 5.4	(k) 22.5	(l) 4.9	(m) 22.9	(n) 0.2	(o) 230	(p) 0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	8.7	40.9	23.2	39.6	23.3	38.4	23.6	28.6	34.0	28.1	33.2	27.7	32.5	2.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.5	23.6	30.6	27.0	23.0	30.1	26.6	22.4	29.8	93.8	34.0	91.3	33.4	89.3	32.5	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.1	(b) 6.0	(c) 5.2	(e) 8.6	(f) 43.3	(g) 2.8	(h) 1.4	(i) 6.6	(j) 44.4	(k) 5.0	(l) 45.2	(m) 3.4	(n) 46.0	(o) 1.3	(p) 47.1		
(a) 7.1	(b) 6.0	(c) 5.2	(e) 4.5	(f) 29.6	(g) 2.0	(h) 0.5	(i) 3.0	(j) 29.9	(k) 1.9	(l) 30.2	(m) 0.7	(n) 30.5	(o) -0.7	(p) 30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	20.0	23.0	27.9	31.0	32.5	32.3	30.2	28.9	29.5	30.0	25.9	21.3
	DBStd	4.56	2.43	3.05	2.60	1.82	1.34	1.49	1.63	1.26	1.41	1.88	2.61	2.64
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	19	11	3	0	0	0	0	0	0	0	0	0	6
	CDD10.0	6472	310	364	554	631	698	670	626	584	584	621	476	352
	CDD18.3	3449	63	134	296	381	439	420	368	326	334	363	226	98
	CDH23.3	42901	587	1326	3479	5104	6224	6029	4731	3761	3968	4577	2267	848
	CDH26.7	22561	139	557	1986	3106	3795	3639	2286	1410	1800	2527	1069	248
	WSAvg	1.8	1.0	1.2	1.5	2.2	3.2	3.0	3.0	2.6	1.8	1.0	0.8	0.8
(15) Precipitation	PrecAvg	400	2	1	1	2	4	35	143	100	81	17	8	0
	PrecMax	1057	18	13	10	60	60	124	465	384	374	223	149	1
	PrecMin	100	0	0	0	0	0	0	21	0	11	0	0	0
	PrecStd	245	5	3	2	11	12	37	122	123	98	50	29	0
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.6	36.2	40.4	42.2	43.2	41.5	37.6	35.5	38.1	39.5	35.7	32.5
		MCWB	18.9	20.4	21.0	22.2	23.9	25.6	27.0	26.4	24.7	22.5	20.4	20.2
	2%	DB	29.6	33.9	38.4	40.5	41.2	39.1	35.9	33.8	36.2	38.1	34.5	30.8
		MCWB	17.8	19.6	20.9	21.7	24.1	26.6	26.6	25.7	25.1	22.4	20.6	19.1
	5%	DB	28.2	31.9	36.8	39.2	39.6	37.8	34.7	32.8	34.8	36.9	33.4	29.4
		MCWB	17.4	19.0	20.7	21.8	24.2	26.7	26.4	25.6	25.3	22.5	20.4	18.4
	10%	DB	26.7	30.1	35.1	37.8	38.1	36.5	33.5	31.8	33.5	35.6	32.3	27.9
		MCWB	16.6	18.0	20.5	21.8	24.7	26.7	26.3	25.6	25.2	22.1	19.7	17.4
	0.4%	WB	22.1	23.7	27.2	27.9	28.9	29.2	28.8	28.6	28.9	27.7	25.6	23.1
		MCDB	27.5	31.4	36.0	36.6	36.9	34.6	32.9	32.5	33.1	34.8	31.6	29.7
	2%	WB	20.6	22.0	24.6	26.3	28.0	28.6	28.2	27.7	27.8	26.6	24.1	21.5
		MCDB	26.2	29.6	33.0	32.8	34.9	33.9	32.2	31.2	32.0	32.6	29.8	27.3
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	19.2	20.8	23.5	25.4	27.5	28.1	27.7	27.0	27.1	25.7	23.0	20.3
		MCDB	25.2	27.8	30.8	31.5	33.5	33.2	31.5	30.2	30.9	31.0	28.2	26.5
	10%	WB	17.9	19.8	22.4	24.6	27.0	27.7	27.2	26.5	26.4	25.0	21.7	19.0
		MCDB	24.2	26.6	29.3	30.6	32.4	32.6	30.7	29.5	30.1	30.2	28.4	25.6
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	13.8	14.1	14.1	11.9	8.7	6.0	5.7	8.3	11.6	12.6	13.2
		MCDBR	14.7	15.1	15.6	15.5	14.6	11.0	7.9	7.4	10.5	12.8	13.0	14.3
		MCWBR	6.7	5.9	4.9	4.2	3.9	2.3	1.6	1.7	2.4	3.9	4.9	6.5
	5% WB	MCDBR	13.7	13.7	14.4	13.7	11.6	8.9	6.6	5.9	7.9	10.5	11.6	13.2
		MCWBR	7.8	6.5	6.5	4.3	3.2	2.4	2.2	2.4	3.0	3.8	5.4	7.4
(40) Clear-Sky Solar Irradiance	taub		0.412	0.434	0.495	0.541	0.598	0.707	0.759	0.685	0.589	0.536	0.517	0.444
	taud		2.123	2.080	1.926	1.854	1.780	1.609	1.548	1.679	1.853	1.882	1.882	2.031
	Ebn at Noon		837	847	812	782	734	653	619	666	724	740	723	784
	Edh at Noon		141	157	192	210	225	265	282	247	204	189	179	150
(44) All-Sky Solar Radiation	RadAvg		4.44	5.28	6.20	6.87	7.00	5.96	4.37	4.37	5.33	5.43	4.51	4.16
	RadStd		0.15	0.15	0.21	0.18	0.21	0.41	0.34	0.29	0.36	0.23	0.21	0.21

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.49	+2.25	+2.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.23	N/A	+1.89	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page